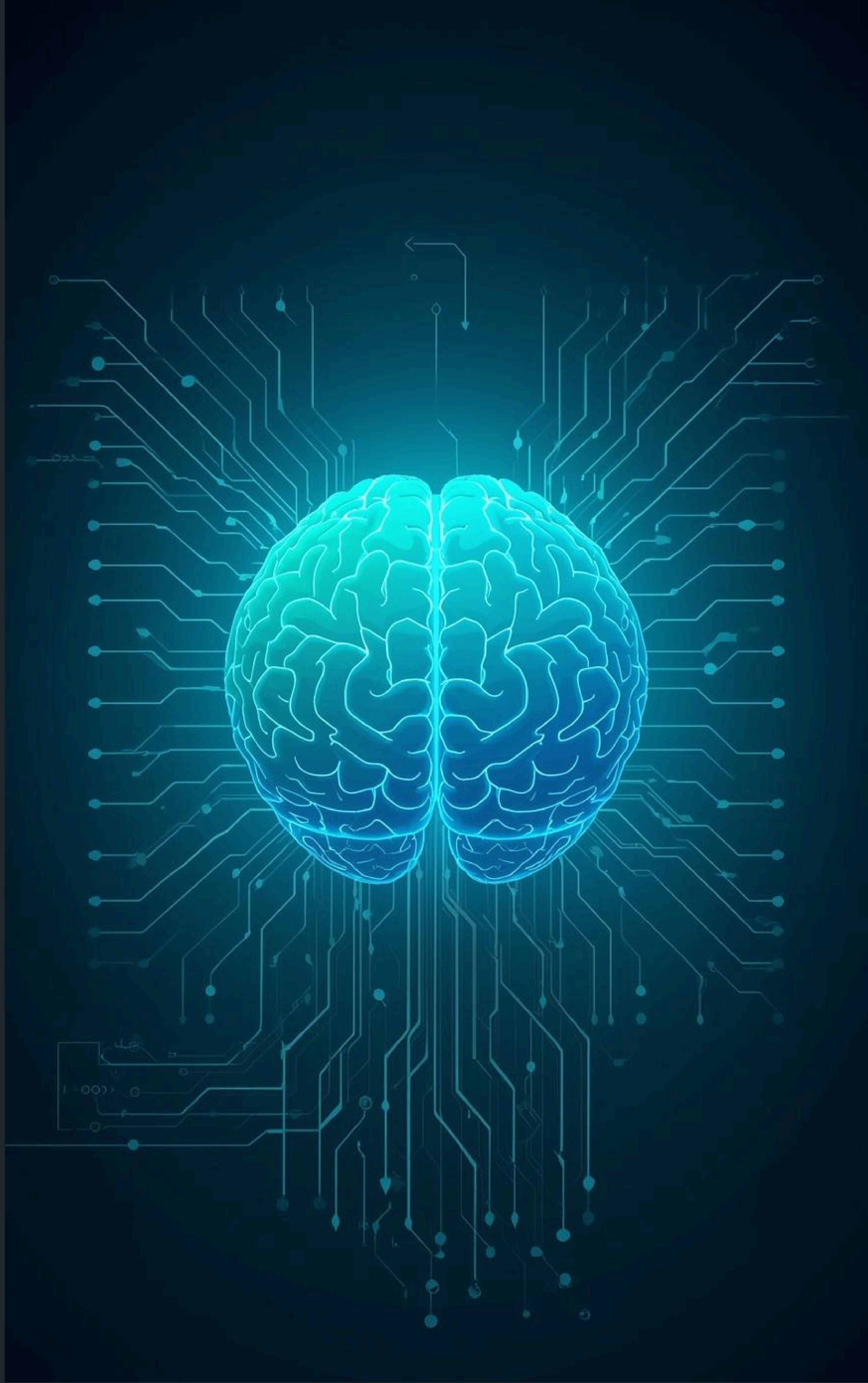
Perceptual Data Structure

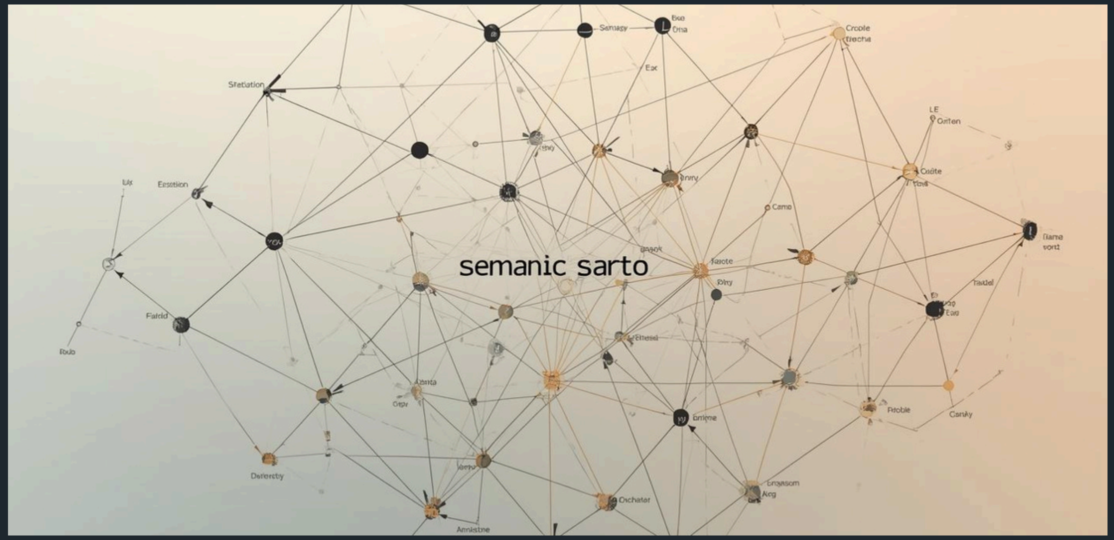


Presented by [Khushi, Avanti]

Core Components

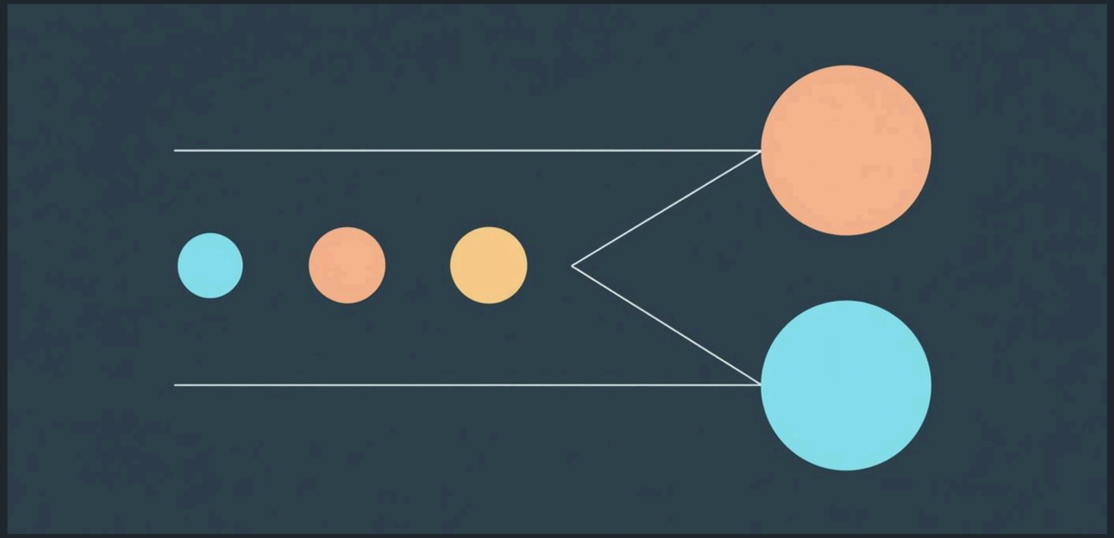
Reflexive System

The reflexive system enables quick responses to stimuli.



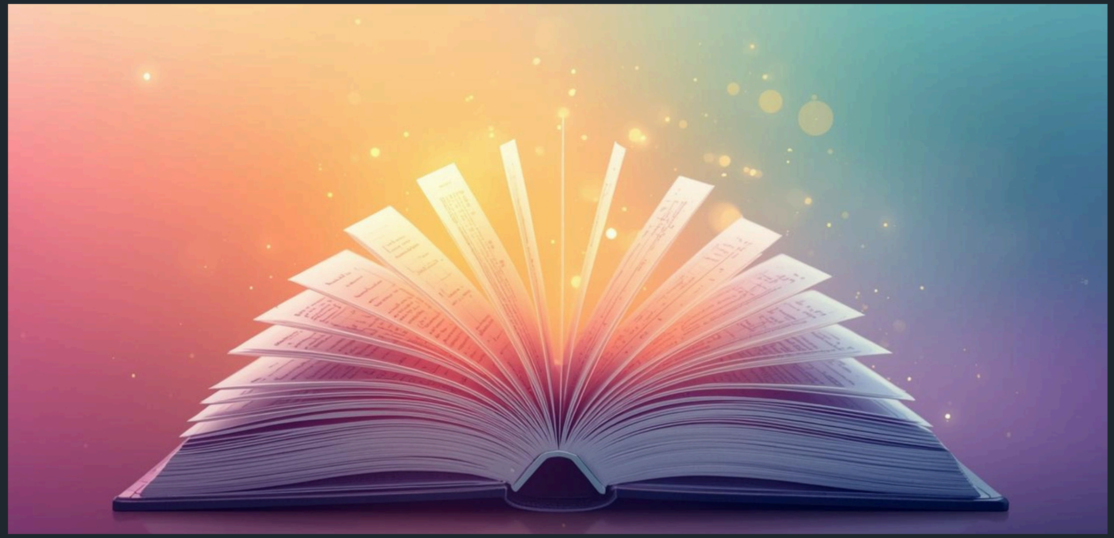
Semantic Model

This model organizes learned knowledge and meanings effectively.



Flask Dashboard

Built an interactive Flask dashboard for this



Memory Database

A memory database organizes and recalls stored information.

Introduction to Learning

Understanding Human Cognitive Evolution

- A child hears a loud dog bark and instantly runs.
- This is a reflex — no memory, no thinking.
- After more experiences, the child learns:
  1) Many barks are harmless → less reaction
  2) Some are threatening → more caution
- Understanding deepens as the child begins to notice:
- Similar situations (close bark vs distant bark)
- Different emotional intensities (scary vs mild)
- Learning shifts from pure instinct to experience + emotion + meaning.

Human learning progresses from instinctual reflexes to deeper memory and reflective processes, shaping our understanding of the world. This model aims to illustrate how our brain learn from new stimuli.

Core Emotional & Decision Variables of the Model

Intensity (Immediate Stimulus Strength)

- Represents how strong or urgent the event feels right now.
- Drives the fast response directly (higher intensity → stronger reflex trigger).

Arousal — Learned Intensity Pattern

- Derived from event intensity.
- Tracks how activating or alerting events of this type typically are.
- Works with valence to form the emotional "circumplex" representation.

Valence (Learned "Good vs Bad" Meaning)

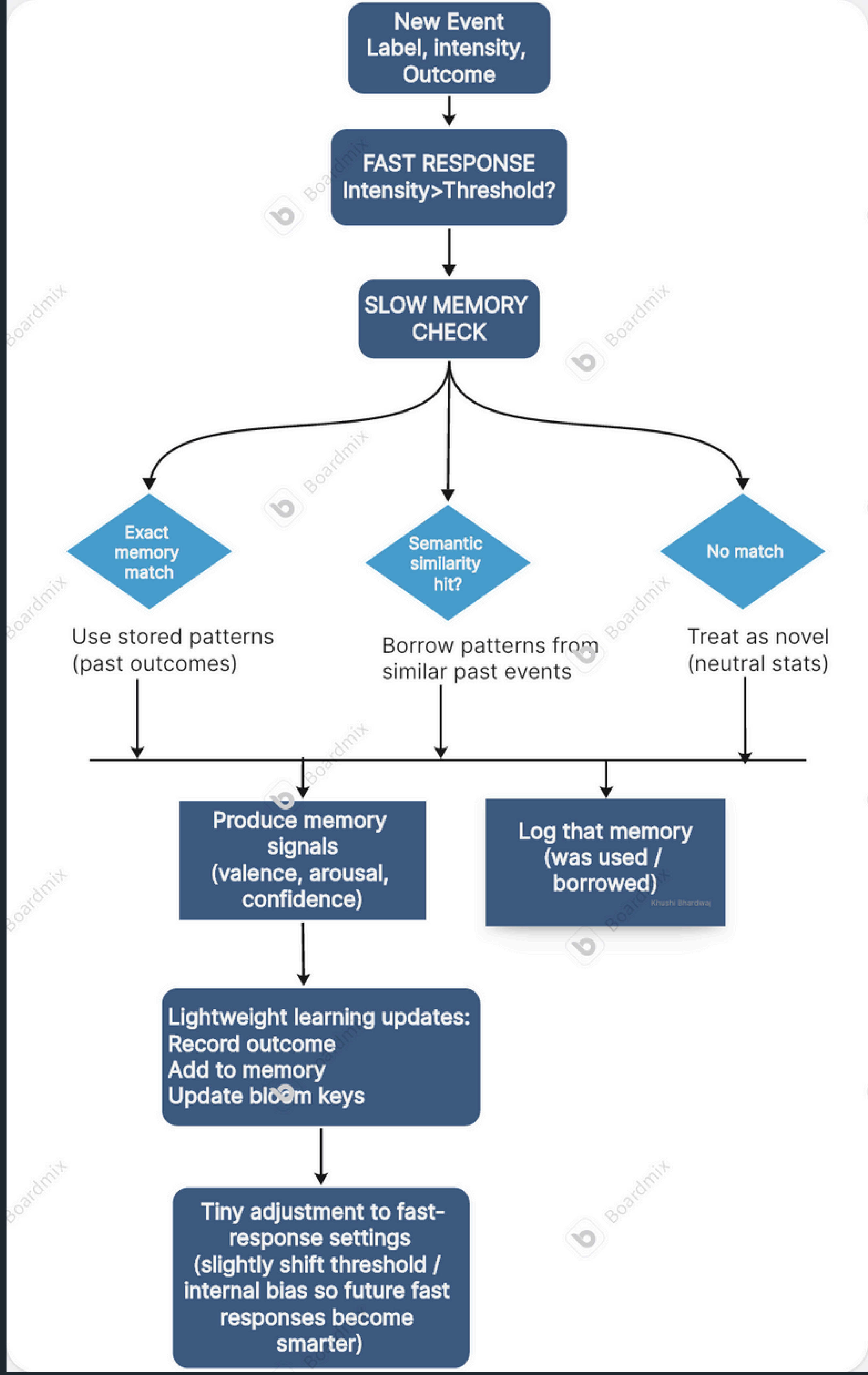
- Computed as: p_safe – p_danger for that label.
- Positive → usually safe, Negative → usually dangerous.
- Encodes emotional meaning and supports semantic borrowing.

Threshold — Adaptive Reflex Sensitivity

- Determines how easily the fast response fires.
- Increases after safe outcomes (calmer reflex).
- Decreases after danger outcomes (more sensitive reflex).

Core Idea of the Model

- Immediate reaction comes first.
- The system produces a fast action using only intensity and its current threshold/policy.
- Memory evaluates the event after the action.
- It checks for an exact match, then semantic similarity, and falls back to "novel" if nothing fits.
- Memory extracts high-level signals.
- Depending on the match type, it produces valence, arousal, and confidence that summarize past experience.
- Outcome is stored to grow experience.
- The system updates memory counts, logs emotional patterns, and records the label for future matching.
- Fast-response rules are lightly adjusted.
- A small update shifts the threshold/bias so future immediate reactions become more informed by experience.



Collision Logic: The Crux of Intelligent Decision-Making

The collision logic mimics how humans use past experiences to understand new situations through pattern recognition and associative memory

1. Semantic Matching

What does this situation mean?
Compares conceptual meaning of current event with stored memories
Uses embeddings to understand linguistic relationships
Example: "Aggressive dog" → matches with "Angry animal" and "Dangerous creature"

2. Emotional Alignment

How does this situation feel?
Compares emotional signatures in valence-arousal space
Measures similarity in both pleasantness and intensity dimensions
Example: High arousal + negative valence → matches with other threatening situations

3. Combined Thresholding

Is this similar enough to matter?
Weighted combination: 60% semantic + 40% emotional
Must exceed collision threshold (0.55) to trigger memory borrowing
Filters out weak or irrelevant associations

Memory Borrowing: Learning from Relevant Experiences

Which Past Experiences Are Used?

- Only the Top 3 Most Similar Memories
- Semantically Related + Emotionally Aligned + Above Similarity Threshold
- Contextually Relevant: Filtered from thousands of memories

Weighted Wisdom Transfer

- Each similar memory contributes based on its similarity strength
- More relevant experiences have greater influence
- Creates composite emotional understanding from multiple sources

Emotional Context Borrowing

- Valence: Borrows pleasantness/unpleasantness assessment
- Arousal: Borrows intensity/activation level
- Enables informed decisions without direct experience

Real-World Example: Novel Event: "Electric fence" (never encountered before)

Borrows From:

- "Hot stove" (65% similar) → High negative valence, high arousal
- "Sharp knife" (58% similar) → Moderate negative valence, high arousal
- "Bee sting" (56% similar) → High negative valence, moderate arousal

Result: System infers danger and caution without ever touching the fence

Use Cases: Dog-Bark Insights

Loud = Danger

A loud dog's bark triggers a **reflexive response** in babies, indicating potential danger.

Soft = Safe

A gentle dog’s presence encourages **exploration** and comfort, promoting positive emotional associations.

Use Cases: Accident Analysis

Car Crash is seen

Car Crash news is heard

Events can be semantically similar yet differ sharply in emotional intensity,seeing a car crash and hearing news about one are not emotionally equivalent.

Future Directions of the Model

• Expanded Emotional Modelling
Add more emotional dimensions beyond valence and arousal—such as fear, curiosity, calmness, or stress level.
This enables finer distinctions between events that feel similar but differ in emotional nuance.

• Memory Consolidation & Forgetting
Introduce a mechanism that gradually strengthens frequently-used memories and fades rarely-used ones.
This prevents memory clutter, keeps reasoning efficient, and mimics human forgetting.

• Context-Aware Interpretation
Incorporate contextual cues like location, time, social setting, or user state.
Similar events can mean different things in different contexts, and the model should adjust its emotional interpretation accordingly.

• Multi-Modal Event Understanding
Combine text labels with additional sensory inputs such as images, audio patterns, or numeric features.
This makes the model more robust—e.g., understanding both the meaning and the visual intensity of an event.

Thank You for Your Attention